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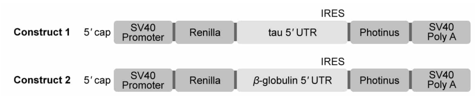
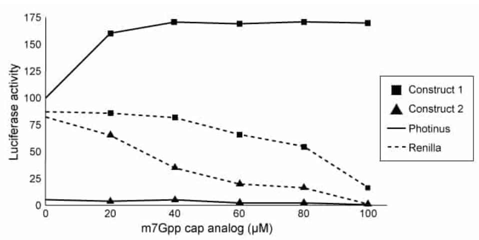


Figure 1 DNA constructs containing *β-globulin* or *tau* gene used to transfect neuroblastoma cells

Luciferase gene expression from the constructs was monitored in the presence of various concentrations of a synthetic m7Gpp cap analog (Figure 2). Translation of *Renilla* luciferase mRNA depended on ribosomal binding to the m7Gpp cap of the mRNA, but translation of *Photinus* luciferase required ribosome recruitment by an IRES in the inserted 5′ UTR.



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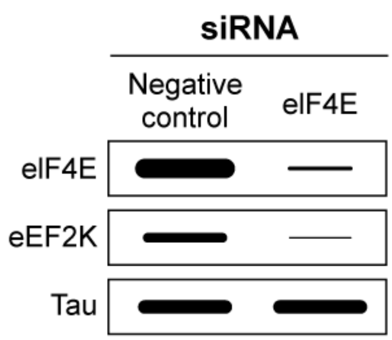


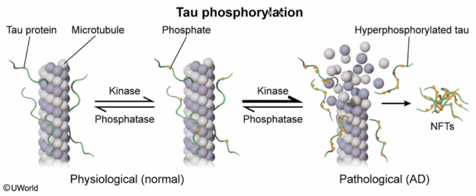
Figure 3 Western blot of eIF4E, eEF2K, and tau extracted from non-AD cells treated with eIF4E siRNA

While analyzing potential therapeutic strategies for AD, researchers would most likely pursue which of the following approaches?

- ✔ A. Phosphatase upregulation [74%]
- B. Kinase upregulation [13%]
- C. Isomerase downregulation [3%]
- D. Synthetase downregulation [8%]

Correct
74% answered correctly

Explanation:



Post-translational modifications occur after protein production. Protein phosphorylation is a reversible post-translational modification that can alter protein conformation, interaction, and function. Kinases are enzymes from the *transferase* class that catalyze the addition of phosphate groups **(phosphorylation)** from ATP or GTP to target proteins. In contrast, **phosphatases** are *hydrolase* enzymes that carry out the catalytic removal of phosphate groups **(dephosphorylation)** from a specific substrate by using water to remove a molecule of inorganic phosphate.

Predicting whether an enzyme will be activated or deactivated based on its phosphorylation status is not possible without experimental confirmation. For different protein targets, phosphorylation can function to increase or decrease protein activity whereas dephosphorylation will generally have the reverse effect. In addition, *hyperphosphorylation* (excess phosphorylation) can interfere with normal protein structure and function.

According to the passage, patients with AD have hyperphosphorylated tau proteins and healthy patients have normally phosphorylated tau proteins. Excessive tau phosphorylation by kinases leads to the formation of neurofibrillary tangles (NFTs), the characteristic phenotype of AD. A likely therapeutic approach would be to either downregulate kinase activity or upregulate phosphatase activity. **Upregulation of phosphatases** would cause the **increased removal of phosphate groups** from hyperphosphorylated tau, leading to **decreased NFT formation and improvement of AD**.

(Choice B) Kinase upregulation would increase the number of NFTs and worsen the AD phenotype.

(Choice C) Enzymes in the isomerase class are responsible for moving functional groups within a single molecule (ie, a reaction in which one substrate generates a single product).

(Choice D) Synthetases, which are part of the ligase class of enzymes, catalyze metabolic synthesis reactions in which the substrates are joined into a more complex compound.

Educational objective:
Phosphorylation and dephosphorylation are post-translational modifications that alter the function or activity of proteins by changing their conformation. Kinases catalyze the transfer of phosphate groups from ATP or GTP to proteins, and phosphatases catalyze the removal of phosphate groups via hydrolysis.

Alzheimer disease (AD) is a progressive condition affecting memory and cognition that can be pathologically characterized by the presence of neurofibrillary tangles (NFTs). NFTs, aggregates of hyperphosphorylated tau proteins that have dissociated from microtubules, are believed to contribute to neuronal degradation and subsequent AD symptoms. In general, ribosomes initiate translation of mRNA by binding the 7-methylguanosine (m7Gpp) cap present at the end of the 5′ untranslated region (UTR) of the mRNA molecule. However, researchers believe that tau protein translation initiation can occur without recognition of the m7Gpp cap. This alternative mechanism, referred to as cap-independent translation, indicates that ribosomes are recruited by internal ribosome entry sites (IRES) generally located in the 5′ UTR of the mRNA sequence.

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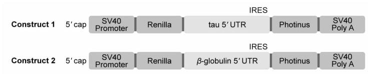
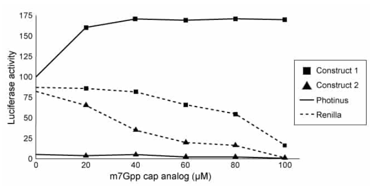


Figure 1 DNA constructs containing *β-globulin* or *tau* gene used to transfect neuroblastoma cells

Luciferase gene expression from the constructs was monitored in the presence of various concentrations of a synthetic m7Gpp cap analog (Figure 2). Translation of *Renilla* luciferase mRNA depended on ribosomal binding to the m7Gpp cap of the mRNA, but translation of *Photinus* luciferase required ribosome recruitment by an IRES in the inserted 5′ UTR.



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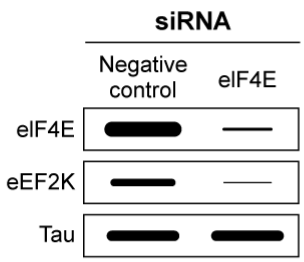


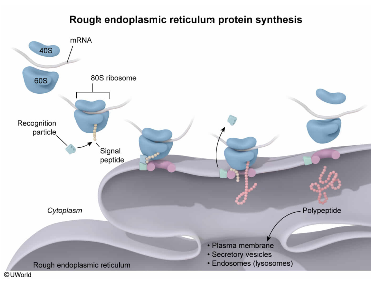
Figure 3 Western blot of eIF4E, eEF2K, and tau extracted from non-AD cells treated with eIF4E siRNA

The cytotoxic protein X.23A induces conformational changes in the large ribosomal subunit, reducing production of secretory but not cytosolic proteins. Treating neuronal AD mouse cells that overexpress tau protein with X.23A would most likely interfere with the binding of:

- A. 70S ribosomes to the nucleolus. [6%]
- ✔ B. 80S ribosomes to the rough endoplasmic reticulum. [62%]
- C. 50S ribosomal subunit to the smooth endoplasmic reticulum. [10%]
- D. 40S ribosomal subunit to the Golgi body. [20%]

Correct
62% answered correctly

Explanation:



Ribosomes are molecular machines that translate mRNA sequences into proteins. All ribosomes are composed of ribosomal proteins and ribosomal RNA, but their structure differs in eukaryotes and prokaryotes:

- **Eukaryotes** produce **80S ribosomes** composed of a **60S large subunit** and a **40S small subunit**.
- **Prokaryotes** produce **70S ribosomes** composed of a **50S large subunit** and a **30S small subunit**.

Ribosomes synthesize proteins in either the "free" state in the cytoplasm or the "bound" state when attached to the rough endoplasmic reticulum (ER). For proteins that enter the ER (eg, secretory proteins), a *signal sequence* within the target mRNA bound to the ribosome is translated into a *signal peptide* that induces transport of the ribosome-mRNA complex to the **rough ER**. The complex synthesizes proteins into the lumen of the rough ER, which then assembles and modifies secretory, lysosomal, and integral membrane proteins.

The cytotoxic (ie, cell-damaging) X.23A protein induces conformational changes in the large ribosomal subunits within neuronal cells of AD mice (*eukaryotic*). According to the question, **X.23A treatment** alters the large (60S) ribosomal subunit and ultimately reduces the production of secretory proteins but not cytosolic proteins. The fact that cytosolic protein production is unaffected suggests that this altered subunit can still bind to the small (40S) subunit and carry out translation. However, the reduced production of secretory proteins suggests an impaired ability of the full 80S ribosome to bind to the rough ER.

(Choices A and C) 70S and 50S ribosomal subunits are found in prokaryotes and are not involved in translation of eukaryotic proteins such as those found in mouse cells. In addition, eukaryotic ribosomes neither bind the nucleolus nor the smooth ER.

(Choice D) After translation, *many* proteins in the rough ER are transported to the Golgi body for further modification and subsequent packaging and distribution to the cell membrane, organelles (eg, lysosomes), and secretory vesicles. However, ribosomes do *not* attach to the Golgi body.

Educational objective:
Ribosomes translate mRNA sequences into proteins in eukaryotes (80S ribosomes, comprising a 60S large subunit and a 40S small subunit) and prokaryotes (70S ribosomes, comprising a 50S large subunit and a 30S small subunit). The ribosome-mRNA complex translocates to the rough endoplasmic reticulum to synthesize secretory, lysosomal, or integral membrane proteins.

Time spent:0 sec
Subject:
Biology

QID:400553
System:
Molecular Biology

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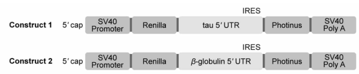


Figure 1 DNA constructs containing *β-globulin* or *tau* gene used to transfect neuroblastoma cells

Luciferase gene expression from the constructs was monitored in the presence of various concentrations of a synthetic m7Gpp cap analog (Figure 2). Translation of *Renilla* luciferase mRNA depended on ribosomal binding to the m7Gpp cap of the mRNA, but translation of *Photinus* luciferase required ribosome recruitment by an IRES in the inserted 5′ UTR.

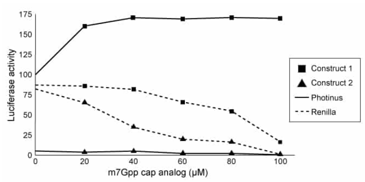


Figure 2 Luciferase activity in neuroblastoma cells exposed to m7Gpp cap analog

In a second study, researchers used siRNA to target initiation factor eIF4E, a peptide translated in a cap-dependent manner that facilitates the binding of ribosomes to the m7Gpp cap. Expression of factor eIF4E, elongation factor eEF2K, and tau protein was assessed in siRNA-treated cells (Figure 3).

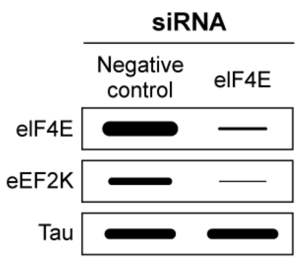


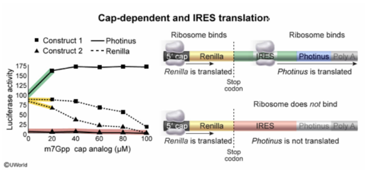
Figure 3 Western blot of eIF4E, eEF2K, and tau extracted from non-AD cells treated with eIF4E siRNA

The data in Figure 2 suggest that ribosomes interact with the:

- A. IRES within Construct 2 only. [4%]
- B. 5′ cap of Construct 1 only. [32%]
- C. 5′ cap of Construct 2 or the IRES within Constructs 1 and 2. [10%]
- ✔ D. IRES within Construct 1 or the 5′ caps of Constructs 1 and 2. [52%]

Correct
52% answered correctly

Explanation:



After **eukaryotic pre-mRNA** is produced, it is converted to mature mRNA to prevent degradation in the cytoplasm:

- The 5′ end of the mRNA is "capped" with a methylated guanosine residue, referred to as the m7Gpp cap.
- A poly-A tail consisting of adenosine nucleotides is added to the 3′ end of mRNA.

In general, ribosomes initiate **cap-dependent translation** by binding the **5′ cap** of **eukaryotic mRNA**. However, the passage introduces a *cap-independent process* in which ribosomes initiate translation by binding an internal ribosome entry site (IRES) within the 5′ UTR of mRNA.

Figure 2 illustrates how increasing concentration of an m7Gpp cap analog competes with mRNA for ribosomal binding. At low concentrations of m7Gpp cap analog, **ribosomes bind to the 5′ cap on Constructs 1 and 2**, and *Renilla* is translated from both constructs. The ribosome then dissociates after reading the stop codon in the *Renilla* mRNA. However, increasing concentrations of cap analog causes *Renilla* protein expression to decrease because the cap analog outcompetes the 5′ caps on mRNA transcripts, reducing ribosomal binding.

Based on the passage, *Photinus* is translated only via a cap-independent process. Figure 2 shows that *Photinus* expression is initially high and rises further despite increasing concentrations of cap analog, indicating that **ribosomes bind the IRES within the *tau* 5′ UTR of Construct 1** to enable *cap-independent* translation of *Photinus*. Therefore, the data indicate that ribosomes can bind the **IRES in construct 1 only** but can bind the **5′ caps of both constructs**.

(Choices A and C) Figure 2 shows that *Photinus* is *not* translated from construct 2, suggesting the ribosomes do *not* bind to the construct 2 IRES.

(Choice B) *Renilla* is translated to some extent in both constructs, suggesting that ribosomes bind to the 5′ caps of both, not just construct 1.

Educational objective:
Eukaryotic translation initiation begins when ribosomes bind the m7Gpp cap at the 5′ end of the mRNA sequence; however, cap-independent processes do exist.

Time spent: 0 sec
Subject: Biology

QID:400554
System: Molecular Biology

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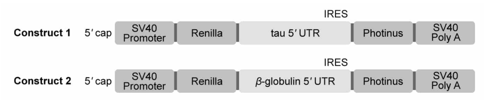

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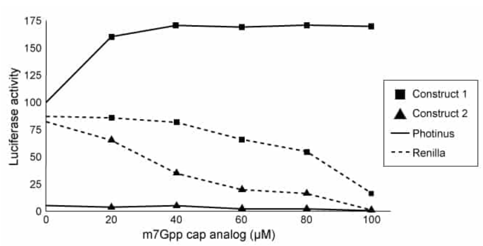

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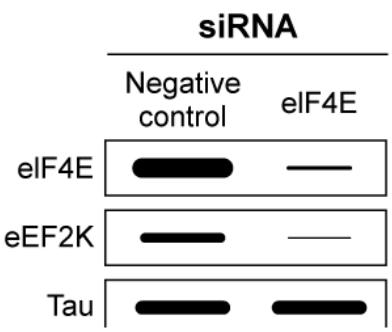


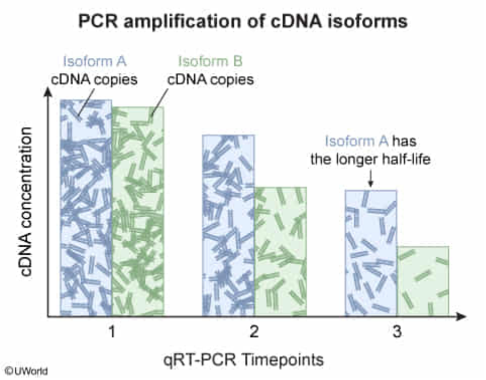
Figure 3 Western blot of eIF4E, eEF2K, and tau extracted from non-AD cells treated with eIF4E siRNA

Tau mRNA has 2 isoforms: v1 and v2. A cell line from AD mice initially maintains constant levels of *tau* v1 and v2 transcripts, and a transcription inhibitor is then added. Of the following, which technique could researchers use to identify the *tau* mRNA isoform with the longer half-life in the cytoplasm of AD mouse cells?

- A. qRT-PCR at a single time point [3%]
- B. Western blot at a single time point [2%]
- ✔ C. qRT-PCR at varying time points [64%]
- D. Western blot at varying time points [29%]

Correct
64% answered correctly

Explanation:



Isoforms of mRNA are different transcripts of the same gene produced by alternative splicing. During **quantitative reverse transcription polymerase chain reaction** (qRT-PCR), the enzyme reverse transcriptase converts mRNA into cDNA, which is then amplified in cycles to yield thousands of copies. The cDNA is initially denatured into single strands using heat, and *forward primers* and *reverse primers* anneal to the denatured cDNA strands so that a heat-stable polymerase (eg, *Taq* polymerase) can elongate the DNA sequence.

Half-life refers to the amount of time needed for the concentration of a substance present in a sample to decrease to half its original value. To identify which *tau* isoform has the longer half-life (ie, lasts longer in the cytoplasm) in the given scenario, the researchers must first convert *tau* v1 mRNA and *tau* v2 mRNA into cDNA. The number of amplified copies (concentration) of tau v1 cDNA can be compared with that of tau v2 cDNA by **performing qRT-PCR at varying time points**. The tau isoform whose concentration decreases *more slowly* over time has the *longer half-life*.

(Choice A) Performing qRT-PCR at only a single time point would yield only one measurement. Although this measurement would show which isoform was present at a higher level at this time point, it is possible that the isoform with the higher concentration was simply maintained at a higher level prior to addition of the inhibitor. Multiple time points are needed to show how quickly the level of each isoform decreases over time.

(Choices B and D) Western blot is used to detect *proteins*, not mRNA. Because the starting material is *tau* v1 and v2 *mRNA*, qRT-PCR must be performed. However, a different technique that also could be employed in place of qRT-PCR is northern blot, which is used to analyze mRNA expression levels.

Educational objective:

Researchers can assess the half-lifes of mRNA isoforms by quantifying mRNA levels via qRT-PCR at varying time points, allowing for comparison of isoform concentrations. The isoform with a smaller decrease in cDNA levels at the final time point has the longer half-life.

Time spent: 0 sec	**QID**:400555
Subject: Biology	**System:** Molecular Biology

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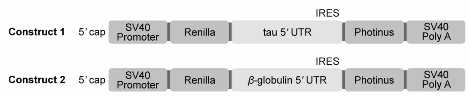

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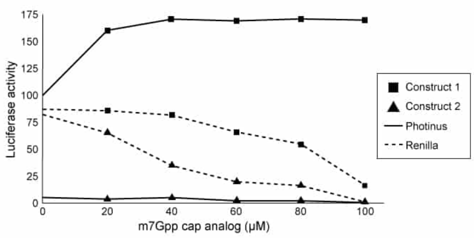

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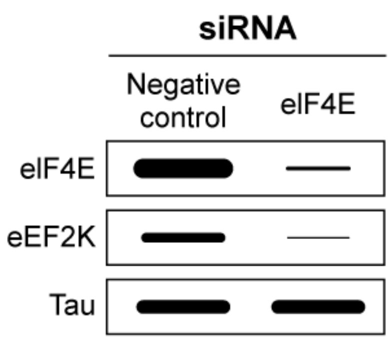


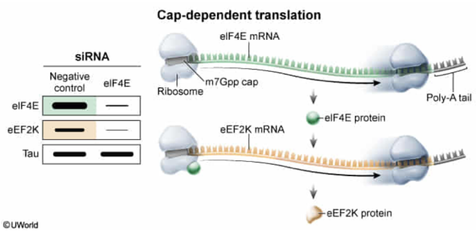
Figure 3 Western blot of eIF4E, eEF2K, and tau extracted from non-AD cells treated with eIF4E siRNA

What conclusion can be made from the results shown in Figure 3?

- A. siRNA only binds the mRNA transcripts of proteins that facilitate IRES-ribosome binding. [13%]
- B. siRNA signals for the degradation of proteins that facilitate elongation of tau proteins. [24%]
- C. eIF4E mRNA is translated via a cap-independent process. [17%]
- ✔ D. eEF2K mRNA is translated via a cap-dependent process. [45%]

Correct
44% answered correctly

Explanation:



Small interfering RNA **(siRNA)** molecules are **short, double-stranded** RNA sequences that **decrease the translation of target proteins**. siRNAs contain complementary sequences that bind to the mRNA of the target protein and signal for its degradation. The western blot in Figure 3 displays the concentrations of initiation factor eIF4E, elongation factor eEF2K, and tau proteins extracted from cells treated with eIF4E-targeting siRNA or a negative control (lacking eIF4E siRNA). As expected, eIF4E siRNA promoted eIF4E mRNA degradation, leading to markedly decreased eIF4E protein concentration.

Paragraph 4 states that eIF4E facilitates ribosome binding to the m7Gpp cap of mRNA, the first step of cap-dependent translation. If eIF4E is absent, a transcript requiring cap-dependent translation cannot be made into a protein. This depletion of eIF4E protein induced by siRNA leads to noticeably reduced expression of elongation factor eEF2K because the eIF4E protein is mostly absent and unable to help the ribosome bind to the m7Gpp cap of eEF2K mRNA. Therefore, **eEF2K mRNA is translated via a cap-dependent process**.

(Choice A) The passage states that eIF4E facilitates ribosome binding to the m7Gpp cap. Because siRNA was able to reduce eIF4E expression, this suggests that siRNA is not limited to RNA transcripts of proteins that facilitate IRES binding.

(Choice B) siRNA signals for the degradation of mRNA molecules (not proteins), which results in less *production* of proteins. Protein *degradation* is *unaffected* by siRNA, but without additional production protein levels will decrease over time. In addition, elongation factors are proteins that facilitate the lengthening of the amino acid chain during translation. The western blot illustrates that the decreased concentration of eEF2K protein does not affect the translation of tau protein.

(Choice C) Paragraph 4 states that eIF4E mRNA is translated in a *cap-dependent* manner.

Educational objective:
Small interfering RNA (siRNA) molecules are short, double-stranded RNA sequences that decrease the translation of target proteins. siRNAs contain complementary sequences that bind to the mRNA of the target protein and signal for its degradation.

Time spent:0 sec
Subject:
Biology

QID:400556
System:
Molecular Biology

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Figure 1 DNA constructs containing *β-globulin* or *tau* gene used to transfect neuroblastoma cells

Luciferase gene expression from the constructs was monitored in the presence of various concentrations of a synthetic m7Gpp cap analog (Figure 2). Translation of *Renilla* luciferase mRNA depended on ribosomal binding to the m7Gpp cap of the mRNA, but translation of *Photinus* luciferase required ribosome recruitment by an IRES in the inserted 5′ UTR.

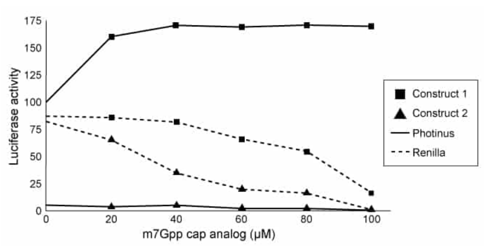


Figure 2 Luciferase activity in neuroblastoma cells exposed to m7Gpp cap analog

In a second study, researchers used siRNA to target initiation factor eIF4E, a peptide translated in a cap-dependent manner that facilitates the binding of ribosomes to the m7Gpp cap. Expression of factor eIF4E, elongation factor eEF2K, and tau protein was assessed in siRNA-treated cells (Figure 3).

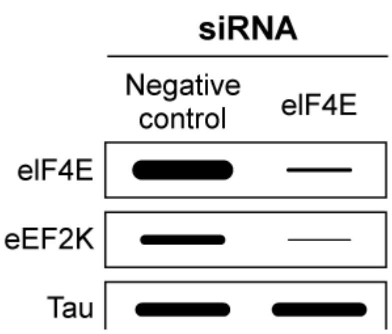


Figure 3 Western blot of eIF4E, eEF2K, and tau extracted from non-AD cells treated with eIF4E siRNA

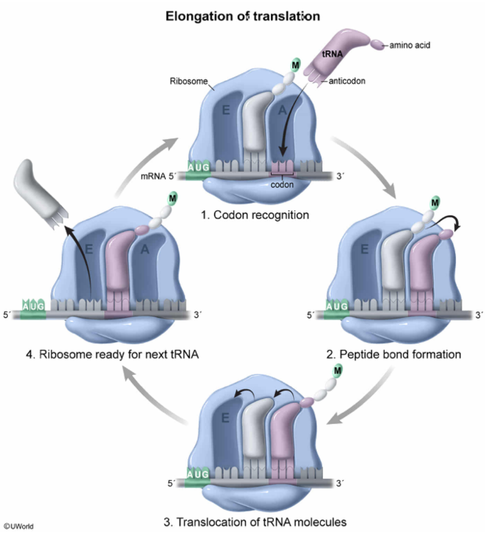
The translation of tau mRNA would most likely involve:

I. The hydrolysis of peptide bonds
II. tRNA molecules carrying amino acids corresponding to codons on the mRNA molecule
III. A ribosomal complex capable of reading the target mRNA in a 3′ to 5′ direction

- A. I only [1%]
- B. II only [68%]
- C. I and II only [21%]
- D. I and III only [9%]

Correct
67% answered correctly

Explanation:



Translation, the energy-requiring process by which ribosomes (with E, P, and A sites) make proteins by reading mRNA, occurs in eukaryotes in the following stages:

1. Initiation: The small 40S ribosomal subunit binds the mRNA 5′ cap and scans the mRNA for a start codon (5′-AUG-3′), which codes for methionine. A transfer RNA (tRNA) molecule "charged" with methionine base pairs with the corresponding AUG codon on mRNA **(Number II)**. Subsequently, the 60S subunit is recruited and binds the initiator tRNA at the P site, marking the formation of the translation complex.

2. Elongation: The ribosome adds amino acids to the polypeptide chain by reading each mRNA codon in a 5′ to 3′ direction. During each elongation step, a new charged tRNA (ie, a tRNA with an ester bond to an amino acid) enters the A site. The ribosome catalyzes *formation* of a peptide bond between the C-terminus of the growing P site peptide and the *free* N-terminus of the A site aminoacyl tRNA, transferring the growing polypeptide chain from the P site to the A site. The ribosome then moves, shifting the newly empty tRNA from the P site to the E site, and the peptidyl-tRNA from the A site to the P site. The E site tRNA is ejected, and a new tRNA enters the A site.

3. Termination: The polypeptide is released from ribosomes when a stop codon (UAA, UAG, or UGA) is detected in the mRNA at the A site, indicating the end of translation. Release factors induce peptidyl transferase to cleave the ester bond between the polypeptide and the final tRNA, causing disassociation of the translation complex.

(Number I) Translation involves peptide bond *formation* (ie, condensation), not hydrolysis. The bond between a charged tRNA and its amino acid is cleaved in this process, but that bond is an ester bond, not a peptide bond.

(Number III) Ribosomes read mRNA in the 5′ to 3′ direction, not the 3′ to 5′ direction.

Educational objective:
Eukaryotic translation involves three stages:

1. Initiation: an initiator transfer RNA (tRNA) charged with methionine is recruited to the start codon.
2. Elongation: The ribosome adds amino acids to the polypeptide chain by reading each mRNA codon in a 5′ to 3′ direction.
3. Termination: A stop codon is read, and release factors induce translation complex dissociation.

Time spent:0 sec
Subject:
Biology

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System:
Molecular Biology